Class	Maximum Permitted Power (mW)	Range (m)
1	100	~100
2	2.5	~10
3	1	~10

Figure 1: Bluetooth classes

Bluetooth Stack	Platform	Availability of Various Protocols of Bluetooth stack			
		RFCOMM	L2CAP	SCO	HCI
PyBlueZ	GNU/Linux	Yes	Yes	Yes	Yes
	Windows XP	Yes	No	No	No
BlueZ	GNU/Linux	Yes	Yes	Yes	Yes
Widcomm	Windows XP	Yes	Yes	Yes	Yes
Java-JSR82	Cross-platform	Yes	Yes	No	No
Series 60 Python	Symbian OS	Yes	No	No	No
Series 60 C++	Symbian OS	Yes	Yes	No	Yes

Figure 2: Various Bluetooth stacks with available protocols

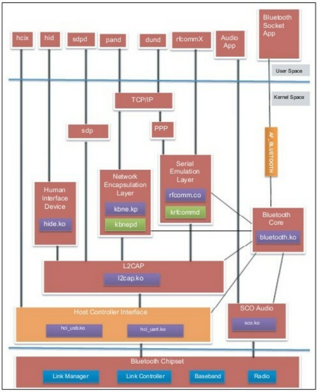

Figure 3: Bluetooth protocol layers mapped to BlueZ kernel modules

- *rfcomm.ko* is responsible for running serial port applications like the terminal. This emulates serial ports over the L2CAP layer. The kernel thread named *krfcommd* is responsible for RFCOMM connections.
- *hidp.ko* implements the HID (human interface device) layer. The user mode daemon *hidd* allows BlueZ to handle input devices like Bluetooth mice.
- *sco.ko* implements the synchronous connection oriented (SCO) layer to handle audio. SCO connections do not specify a channel to connect to a remote host; only the host address is specified.

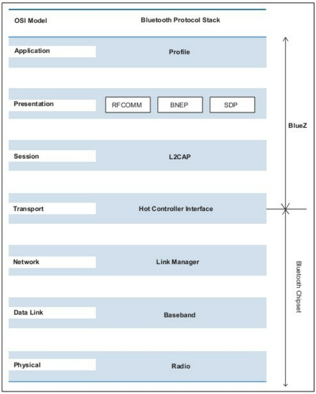

Figure 4: Bluetooth protocol stack vs the OSI model

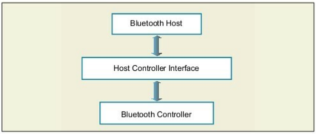

Figure 5: The Bluetooth protocol stack

The Bluetooth protocol stack vs the OSI model

The reason for comparing the Bluetooth protocol stack with the Open Systems Interconnection (OSI) model is to understand the former, better. Knowledge of the OSI model is assumed.

The radio, baseband and link manager layer are the hardware part of the Bluetooth architecture which is implemented on the Bluetooth chipset. These layers roughly map to the physical, data link and network layers in the OSI model. The host control interface (HCI) maps to the transport layer, and transports data between the L2CAP layer and the Bluetooth chipset. The Logical Link Control and Adaptation Protocol (L2CAP) maps to the session layer. Serial port emulation using radio frequency communication (RFCOMM), Ethernet emulation using Bluetooth Network Encapsulation Protocol (BNEP) and the Service Discovery Protocol (SDP) are part of the feature-rich presentation layer. At the top of the stack reside various application environments called profiles. Since the radio, baseband and link manager are usually part of